

Miguel Vilhena

Frontend engineer, fullstack when it counts, deep in AI

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I take products from a blank page to 30 countries. I joined Rewards from IKEA Family as a one-country pilot and have carried it through every market launch since, the first points-earning mechanic in IKEA Family's forty-year history and now live to a base of over 200 million members. Eleven years building product platforms across loyalty, enterprise networking, healthcare and public transport. Founded and exited two companies alongside full-time work. I build with AI daily and run my own local model infrastructure.

SELECTED WORK

PowerByJS

Case study miguelvilhena.com/powerbyjs

Brand, site and booking flow for a Haarlem strength studio, designed and built end to end. Full case study, including the problem, the decisions and the result, at the link above.

EXPERIENCE

IKEA Full Stack Software Engineer

Jul 2023 – Present · Amsterdam, NL

Joined Rewards from IKEA Family as a Portugal pilot and carried it through every market launch since, to 30 countries. The first points-earning mechanic in IKEA Family's forty-year history, live to a base of over 200 million members.

- Rebuilt the claim and redeem flow, the path every reward in the programme goes through, and took offer claims up sharply.
- Built the launch system the team used to take the whole loyalty programme into every new geography, 30 markets in all. What used to need me now needs a checklist.
- Led the V2 initiative end to end: the micro-frontend rebuild, and the migration onto the upstream's new canonical model through major breaking changes, without dropping a market.
- Worked across the stack, Preact and TypeScript on the front, Node, Firestore and GCP behind, and wrote the architecture decisions down before they got built.

Preact · TypeScript · Node · Express · Firestore · GCP · Terraform · Terragrunt · GitHub Actions · Contentful · Optimizely · Agentic AI

Expereo Software Engineer

Mar 2022 – Jul 2023 · Amsterdam, NL

Shipped the world's first B2B Intelligent Internet Platform, and fixed how the team delivered it by taking on Scrum Master alongside the code.

- Built the frontend of the platform, giving enterprise customers a single view of global network performance.
- Removed the drag of a growing monorepo by standardising it on NX, Tailwind, Twin.Macro and generated Codegen clients, so types and styling stayed consistent as the team grew.

React · TypeScript · NX · Apollo · GraphQL · MongoDB · AWS Cognito · Tailwind · GitLab CI

INFO Software Engineer

Oct 2021 – Mar 2022 · Amsterdam, NL

Built NFC unlocking into the OV-fiets operator app at NS. The same mechanism now lets riders open bikes themselves, on a fleet of 22,000 nationwide.

- Added NFC unlocking to the operator app, so staff opened a bike by holding their NS handset against its e-lock instead of going through the back office.
- That mechanism grew into the self-service unlock riders use today with their own travel card, which took the operator out of the rental entirely.

React Native · TypeScript · REST APIs · iOS & Android

Prime IT Software Engineer, client Aquasis / AGS

Mar 2021 – Sep 2021 · Lisbon, PT

Rescued a low-code water-monitoring platform that had outgrown its foundations, rebuilding it in React to keep Portugal's water networks observable in real time.

- Rebuilt the platform from scratch in React after it hit the ceiling of its original OutSystems implementation, delivering version 2 inside seven months.
- Redesigned the UX/UI end to end, then built the frontend against it, turning a dense monitoring tool into something usable under pressure.

React · .NET Core · SCSS · OutSystems · Figma

Squad IT Software Engineer

Feb 2020 – Mar 2021 · Lisbon, PT

Found and permanently fixed a defect that was failing to register every automatic prescription issued between 22:00 and midnight, every day, at one of Portugal's largest private hospital groups, closing a nightly gap in patient care. Then built the group's next system from scratch, and modernised the legacy estate without interrupting clinical operations.

React · OutSystems · SQL

Grupo ANF IT Risk Engineer

Jan 2018 – Feb 2020 · Lisbon, PT

Built the system that let a multi-business group see its own IT risk for the first time, standing up the risk function from nothing alongside the CSO.

Deloitte Software Engineer & IT Auditor

Feb 2015 – Jan 2018 · Lisbon, PT

Audited where the risk was in Portugal's largest companies, then shipped the CRM software that closed it.

- Built and extended clients' CRM systems, writing new services and wrapping legacy ones so decades-old estates could be reached by modern software.
- Delivered across a portfolio of top-tier clients including EDP, Repsol, Iberdrola, REN, NOS, FNAC and Sonangol.

CRM Development · Java · SQL · IT Audit

FOUNDED COMPANIES

Built and run alongside full-time work.

UnicornCreation.net Co-Founder & CTO

2020 – 2021 · Belgium

Co-founded a consulting studio that turned EU digitalisation funding into shipped software for Belgian SMEs, and owned every technical decision as CTO. Exited by selling my stake to the other co-founder.

Respets Founder & Software Engineer

2016 – 2018

Built, ran and sold a transatlantic e-commerce brand single-handedly, writing every line of the storefront and owning the P&L behind it. Sold it to an online competitor in a successful exit, after proving the model rather than raising against it.

BUILDING WITH AI

I run a private AI development environment on a Mac Studio at home: a local open-weight LLM served on Apple silicon, an agent orchestrator, and a chat interface I can reach from anywhere over Tailscale, sandboxed to a non-root user and a constrained workspace.

I use it to take an idea from a loose prompt to a structured technical spec, then to designs, code, and automated quality gates, with sub-agents owning each stage. Agentic AI is also part of my day-to-day work at IKEA.

EDUCATION

NOVA IMS, Universidade Nova de Lisboa · BSc, Information Systems and Technology

2011 – 2015

Named to the NOVA IMS Academic Merit Board for 2013/2014, top five students on the course.

Boot.dev · Backend Development and Computer Science

ongoing

FLAG Lisbon · UX/UI and Front-End Development

2020 · 19/20

Lisbon School of Design · Brand Design

2019 · 18/20

LANGUAGES

Portuguese (native) · English (fluent) · Spanish (advanced) · French (basic)